2024 Ph H2 Q13

Section: Electricity

Topic: Oscilloscope & AC

Summary of Question

A signal generator drives an oscilloscope. The trace and the selected Y-gain are shown; the frequency of the signal is 250 Hz. The timebase setting is not shown.

Tasks: (a) Define alternating current (AC). (b) Determine the rms voltage from the trace and Y-gain. (c) Determine the timebase setting.

(a) What is meant by AC?

Answer: An alternating current is an electric current that reverses direction and changes its instantaneous value periodically with time.

(b) rms voltage from the trace

From the screen: the peak amplitude is 3.0 divisions and the Y-gain is set to 5.0 V/div.

Peak voltage: V_{peak} = (3.0 div) × (5.0 V/div) = 15 V.

rms voltage: $V_{rms} = V_{peak}/\sqrt{2} = 15/\sqrt{2} = 10.6$ V.

(c) Timebase setting

Given frequency $f = 250$ Hz → period $T = 1/f = 4.0 \times 10^{-3}$ s.

From the trace: one complete cycle occupies 4 divisions.

Timebase = T / (divisions per cycle) = (4.0 ms) / 4 = 1.0 ms/div.

Final answer: 1.0 ms/div

Revision Tips

Oscilloscope: V_{peak} from vertical divisions × Y-gain; $V_{rms} = V_{peak}/\sqrt{2}$ for a sine wave.

Timebase = (period) / (horizontal divisions per cycle).

Always quote units and use the given frequency to cross-check your timebase.